



Sergio Ballesteros

Backend Engineer

Backend Engineer specialized in microservices, data processing and AI integration. I design and build production systems used in commercial operations in the energy sector, combining Java, Spring Boot, Python and cloud architecture.

- ✉ sergioabch@icloud.com
- in linkedin.com/in/sergio-ballesteros-dev
- github.com/sergioballesteros-vd

PORTFOLIO

sergioballesteros.dev

Madrid, Spain

Core Focus

Java · Spring Boot · AWS · Kubernetes · AI Integration · Data Engineering · Automation

Experience

VDenergy · Backend Engineer

Mar 2023 - Present · Madrid, Spain

Build and operate backend services, automation workflows and AI integrations used in commercial and operational processes.

- Designed, developed and maintained Java microservices with Spring Boot used in production for critical business processes.
- Designed REST APIs for pricing, scoring and operational workflows that needed predictable behavior under load.
- Designed ETL pipelines capable of processing more than 900,000 records, automating validation, cleanup and enrichment.
- Integrated OpenAI and Anthropic into production workflows to automate scoring, document extraction and commercial insights.
- Deployed and operated services on AWS, Docker, Kubernetes and GitLab CI/CD.
- Automated internal processes, reducing analysis time by up to 80% in target workflows.

Impact

- Automated internal workflows with a reduction in analysis time of up to 80%.
- Processed more than 900,000 records through ETL pipelines with validation, cleanup and enrichment.
- Built OCR + LLM extraction flows with more than 98% accuracy in the target workflow.
- Shipped and maintained production microservices used in commercial operations.

Selected Systems

OCM Brain

AI Platform

Service-backed AI layer for lead scoring, consent freshness checks and sales-facing insight generation.

Energy Simulator

Internal Platform

Shared simulation backend for pricing scenarios, tariff comparisons and reusable commercial analysis.

Smart OCR

AI Automation

OCR + LLM extraction pipeline with structured validation and more than 98% accuracy on the target workflow.

Data Processing

Data Engineering

High-volume import, cleanup and ETL workflows with more than 900,000 records processed.

Technology

BACKEND

Java, Spring Boot, Python, FastAPI, REST APIs

CLOUD

AWS, Docker, Kubernetes, Git, GitLab CI/CD

DATABASES

PostgreSQL, MySQL, Redis, SQL

AI

OpenAI, Anthropic, OCR, LLMs, Pydantic

FRONTEND

Next.js, TypeScript

Public Repositories

smart-document-extractor, async-task-dashboard, vtx-hexagonal-template.